

A Transient Groundwater Model for the Heathcote Valley

Objectives

Modify the steady-state groundwater model of the Heathcote Valley from last week's tutorial for a transient problem. You will need to use this model for the Assignment.

Resources

These instructions and this week's lectures on using Model Muse to set up transient MODFLOW models. A Python script for analysing the output of wells.

Instructions

Open the Heathcote Valley model you developed last week and review the tasks you went through.

Transient simulation set up

We will start by setting up the model packages that support a transient simulation. The nominal simulation period will be an additional 1 year on top of the steady-state simulation. The following steps need to be completed.

1. From the Model > MODFLOW Time menu, add a second stress period, a transient step lasting 1 year with numerous time steps (recommend a first time step of 3155 and a multiplier of 1.1).
2. From the Object > Show or Hide Objects menu, select relevant objects and modify the time extent of the rainfall and ocean recharge boundary conditions so that they match the new end date of the model.
3. **You will need to assign a specific storage. Use $9 \times 10^{-5} \text{ m}^{-1}$.**

Set up the model to include the transient stress period.

Pumping of Well BY20/0083

A water supply well has been drilled into the main aquifer (lowermost layer) and has been producing at 40 L/s for one year. Its location is marked by a red circle on the annotated topo map (ensure you select the correct well).

1. From the Model > MODFLOW Packages and Programs menu, turn on the WEL package.
2. Add a point object at the location of the pumped well and ensure its Z-coordinates are specified to only contain the bottom layer.
3. Activate the WEL package for this object and assign its (DIRECT) pumping rate to 40 L/s (make sure to convert your units and to use the correct sign).

Set up the model to include a pumped well and then run the simulation.

Visualising the Cone of Depression

Drawdown is a change in head relative to some reference time. This is usually the pre-exploitation era, which for us is the end of the steady state simulation. Given simulated head output at two different times, we can take their difference to look at contours of drawdown around a well, also called the Cone of Depression.

1. From the File > Import > Model Results menu, load the model output and make sure to check both the FIRST and LAST results in the list of output times.

- From the Data > Data Visualization menu, look at 0.1 m spaced contours of head at the first and last time steps in the aquifer layer. You should get something like the two plots below.

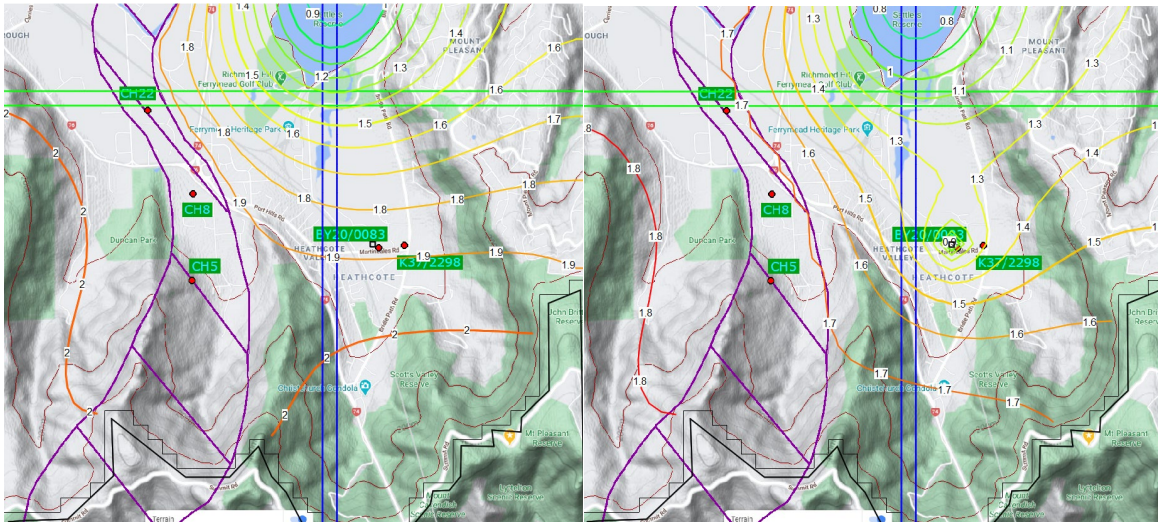


Figure: Lower aquifer head before (left) and after (right) pumping. You can turn off the model grid in View > Show or Hide 2-D Grid > Show Active Edge.

To draw the cone of depression, we need to create a New DataSet that is the difference between two existing datasets (the before/after pumping head distributions above).

- In the Data > Edit Data Sets menu, select the Add button and give your NewDataSet a sensible name.
- Click the Edit Formula button to open a new dialog. Unfold the Data Sets > Optional > Model Results > Layer Data and study the options. These are head output for different stress periods (P^*), for different time steps (S^*) and for different layers (L^*).
- Compute the Drawdown in the lowermost layer by entering/selecting

$$\text{Head}_{P2_S^{**}}_{L3} - \text{Head}_{P1_S01}_{L3}$$

where $**$ is the final time step in your model (it was 73 for me).

- You should now be able to plot contours of drawdown from the Data > Data Visualization > Contour Data menu (your new Data Set is right down the bottom under User Defined).

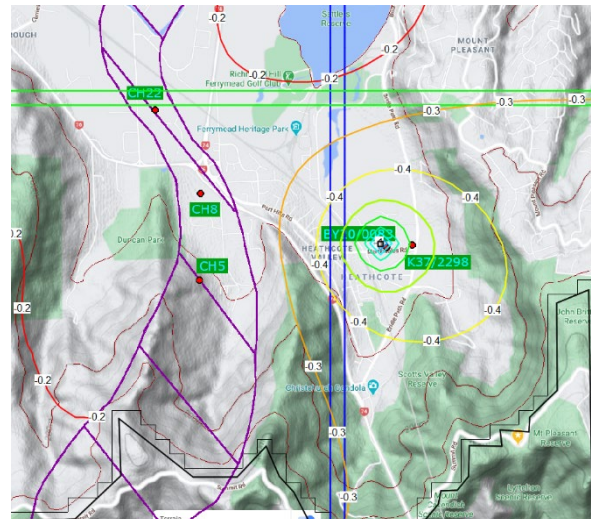


Figure: Contours of drawdown in the lower most aquifer layer around simulated well BY20/0083.

Particle Tracking

Particle tracks are a nice way to view where the water we are pumping is coming from. We'll use backward time particle tracking to figure out where the pumped water is coming from.

- In the Model > MODFLOW Packages and Programs menu, turn on MODPATH. Make sure to set the Tracking direction to Backward, and the Reference time as the final simulation time.

2. Open our Well object and assign it some particles. Make sure to spread them around as a Cylinder and to set the Release time as the final simulation time.
3. Rerun the model and then visualise the particle tracks via the Data > Data Visualization > MODPATH Pathlines dialog.

The particle tracking is often quite fussy. If you are having difficulty getting the particles released, set the MODFLOW Time, transient time settings to the recommended values in the [Transient simulation set up](#) section above.

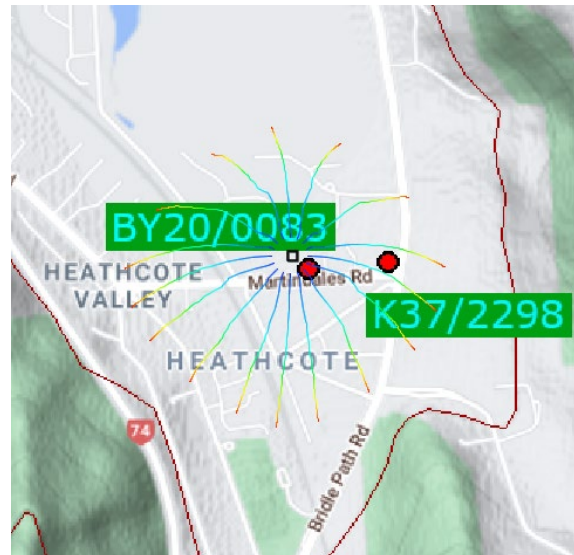


Figure: One-year particle tracks constructed around simulated well BY20/0083 defining a source protection region.

Predicting Drawdown over Time

One of the great things about groundwater models (most numerical models!) is they can be used to predict the future. In MODFLOW, that amounts to using the observation package to generate a time series output of evolving head at specific locations.

We are interested to understand drawdown in the model block containing the well. Note, this is *not* the same as the drawdown inside the well itself (which has a much smaller diameter than the block it sits in: 0.6 m vs. 100 m). Instead, it is a measure of drawdown in the cone of depression *averaged* across the block area.

To output head changes at the well, do the following:

1. In the Model > MODFLOW Packages and Programs menu, turn on the OBS package.
2. In the Well object, activate the OBS package. Ensure you assign this observation a sensible name and that you have checked the Head observation box.
3. Rerun the model.

These changes ought to generate a file `heathcote.ob_gw_out_head.csv` or something similar. Open the file and check you understand its contents.

Plot the contents of the file using Python.

1. In your preferred Python IDE (Wing, PyCharm, Visual Studio Code) and change the working directory to where your MODFLOW has been developed.
2. Open the Python file called `drawdown_analysis.py` that is linked on LEARN

3. Run the script to obtain the plot below, changing the **name of targeted MODFLOW output file** if necessary.
4. Note, you may want to change your minimum time step to something smaller to investigate the early time behaviour.

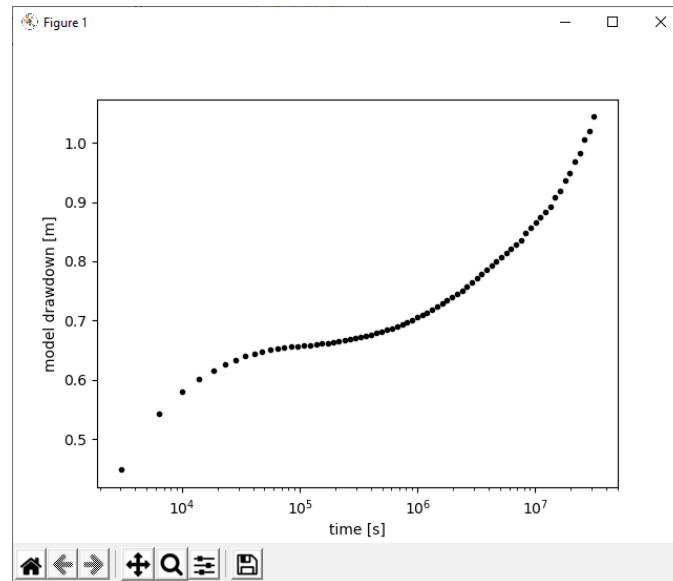


Figure: Drawdown vs. time in the model cell containing well BY20/0083.

What kind of aquifer does this curve imply? What do the early, middle and late sections of the curve reflect?